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Anomaly Detection and Classification in Cellular Networks Using Automatic Labeling Technique for Applying Supervised Learning

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Abstract

Anomaly Detection (AD) is a promising new approach for quality control in e.g. operational telecommunications and data networks. In this paper we have applied Supervised Machine Learning (SML) to a set of long term observation time series from a Cellular/Wireless network. We have shown that periodically collected Key Performance Indicators (KPIs) can be analyzed by supervised ML. Generally, the network creates a new big data periodically when different KPIs from e.g. all the cells (sectors of each 2G/3G/4G/5G base station) are output to a remote Database. We have applied a single class support vector machine in the first phase to find out outliers in range based KPI values. Then LSTM RNN (Recurrent Neural Network) is used for deeper understanding of their behavior over time. Both profile based KPIs and range based KPIs are used to filter out the FP (False Positive) or FN (False Negative) anomaly candidates. In this study, we have applied a novel approach to automatically label the huge data into a supervised training set. This is possible when the meaning of major KPIs is well understood. Both a time series profile based prediction and a logical combination of acceptable value ranges (Min/Max) are used for Anomaly Filtering (AF). A Min or a Max condition is omitted in a single threshold case. AF is used both for AD and for automatic labelling of the training set for ML. Automated labelling with AF performed well also for any large dataset. The pure time series graph profile based KPIs without applicable limits were not used for labelling nor for AF. This technique gave us better results than unsupervised learning based AD. Our enhanced supervised AD decreased the number of FP anomalies from 33 to 0, while the total anomalies decreased from 35 uncertain cases to 2 TP (True Positive), 0 FN. Finally, KNN algorithm is used to classify test data sets. Our proposed method seems to solve several major problems in the field of Cellular/ Wireless, Fixed, [Packet (e.g. IP)] Data Networks as well as within related network side and user equipment. Automation in general, including medical/ any critical systems and equipment is another possible application domain. Automated labelling with AF performed well also for any large dataset.

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1. Introduction

Cellular / Wireless / Fixed Telecommunications or Data Network Operators / Carriers face the challenge to minimize technical problems experienced by their customers, subscribers. Examples of such problems are failing setup of a packet data session (including VOIP, VoLTE) or a basic circuit switched voice call, noisy video/voice quality possibly with interruptions, slow/variable download/ upload speed or unintentionally lost connections like dropped voice/ video calls. It is hard to detect a fault having neither any alarm nor any related downtime. We call them *Hiding HW-Faults / SW-Bugs*. They are not detected by the self-diagnostics of the Network Element (NE). The NE cannot have diagnostics for all possible unexpected fault scenarios without becoming prohibitively complex, slow to develop and expensive. Faults in passive components without any intelligence are also typical cases (cables, connectors). Luckily any advanced NE contains statistical Performance Measurement (PM) counters that the NE periodically uploads to a remote OSS Database in the O&M center. In this paper we cover detection of also Hiding Faults/Bugs using both the raw PM counter values and values of derived formulas - Key Performance Indicators (KPI) - as input time series. Examples of 4G KPIs[15].

2. Related Works

Liu et al. [1] introduced a tool, which explains the AD process using ML and labelling the anomalies manually. Their assumption is that anomalies are rare and training data labelling can thus be updated manually and regularly with their user-friendly tool.

Ciocarlie, G. F. et al. [14] explained in their studies that anomalies can be detected using KPI measurements. They have compared selected KPIs to measure the degradation of cell level measurements. This method expects a slider to mark the real anomalies by the operator. Ciocarlie, G. F. et al. [4,5] presented another study about cell AD using ensemble method.

In our study, we combined profile based and range based anomaly detection methods into a proposed Anomaly Filter (AF). To the best of our knowledge, this is the first study to combine both of the methods. We propose additionally an automatic labelling technique. It completes full automation of the training dataset generation. AF reduced FP anomalies compared to existing unsupervised learning making supervised ML feasible.

3. System Design

We searched for anomaly detection applications to Cellular / Wireless networks. We concentrated on most related works [3, 6, 7, 11]. As a major difference, we used our AF also for automatic labelling of the training set. Visual inspection showed that there was not a single false label at all. Our designed SW-modules were able to process the big data set.

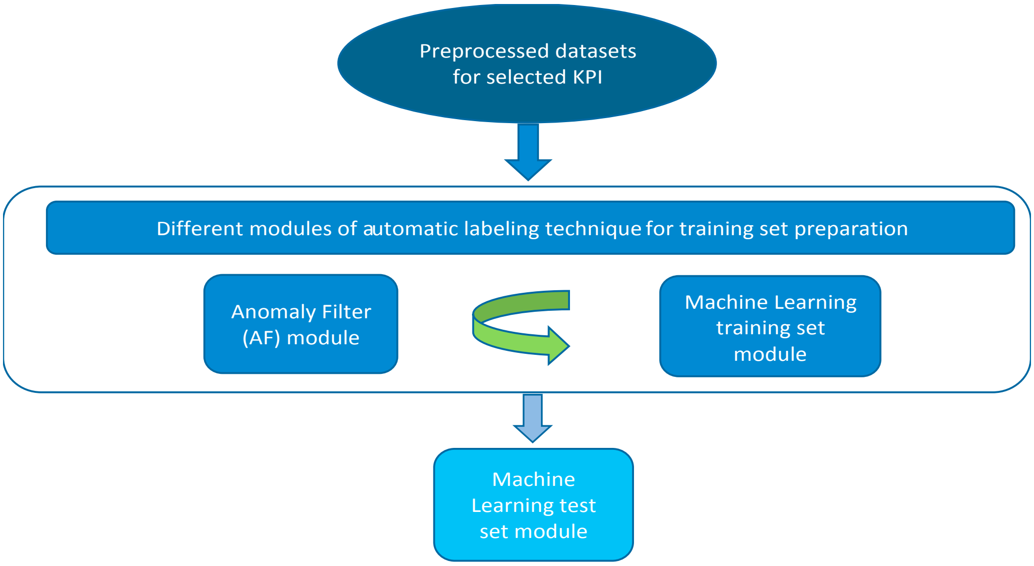


Figure 1: Automatic labelling algorithm modules for supervised machine learning training set generation

Figure 1 demonstrates the design of the process we describe in this paper. It is separated into 3 main modules. The ML training set module works with the AF module to find out the final anomaly candidates and to automatically labelling the training set. This supervised approach classifies the test data sets using KNN (K-nearest neighbour) algorithm with nearest 3 neighbour. We achieved 98.82% accuracy after AF was applied. We used unsupervised ML algorithm to predict the set of range based anomalies (outliers) using one class SVM (Support Vector Machine). LSTM (Long Short Term Memory) based RNN (Recurrent Neural Network) deep learning model is used to predict profile based KPI values, and indirectly the related set of profile based anomalies. The intersection of these two sets contains our final set of anomalies.

4. Pre-processing of KPI data

A single cell or a few, randomly selected, cannot be expected to produce any repeating daily profile pattern. Especially low traffic cells out of hot spots have a lot of randomness in their daily profile of events. There is no obvious reason for people to repeat their cellphone activities at the same time on the next day. In addition local outdoor / indoor events or a traffic accident causing a traffic jam on a specific day in a specific time interval do not keep repeating the next day. Therefore, we selected cells from a medium size town and summed up the corresponding event counters hoping to see a mostly repeating daily profile [9].

We had no idea if there would be even a single real anomaly in the data. For this reason, we selected a locally special set of days, the fasting month of Ramadan followed by the first two days of festival celebration. We included only the weekdays Mon-Fri during Ramadan to get a more regular daily pattern, followed by a single official half-working day (Thu in 2018), followed by two first holidays (Fri-Sat) when the fasting is over and people celebrate their festival. The last 2-3 days could be expected to break the regular daily profile and create anomalies compared to the preceding 21 fasting weekdays. As we hoped and expected, we had success. Selection of this period enabled testing of our ideas.

5. Comparison of Three Methods for Anomaly Detection

5.1 Range Based Anomaly Detection

A logical combination of one or more acceptable value ranges is used for AD.

(Min_OK <= KPI <= Max_OK) *(Min_OK / Max_OK condition is omitted in case of a single threshold)*
There are 4 system parameters per KPI range: The limits (*min_OK, max_OK*) and an on/off switch per limit (*min_OK_used, max_OK_used*).

Let's assume the acceptable range is **[0, 2]** for a given KPI in this example (see **[15, 17]** for examples of KPI)

Network Element	YYYY-MM-DD	Hour (HH)	KPI-Value	Anomaly?
…	…	…	…	…
QRS1	2018-05-07	17	0.3	No
TUV2	2018-05-07	17	**3.7**	**Yes**
…	…	…	…	
XYZ9	2018-05-07	23	0.7	No

Table 1: Example of acceptable range based anomaly detection (0<= KPI <= 2)

Pros: +Also Old ongoing Anomalies get detected.

Cons:

- Too few samples may produce many false positive detections. This is common in time intervals of low activity unless a min number of samples is additionally required. However, with such a condition false negative would occur repeatedly for a problem NE with too few samples in every interval. The anomaly would repeatedly remain undetected as a FN.

- Detection of impulse-like (peak/dip) or step-like changes in KPIs without clear limits (CS_Traffic [Erl], Data_Throughput_DL_FL_[Mb/s], Data_Volume_UL_RL [GB], ..). It would require tuning of linear / non-linear (median) digital filters considering a time window over multiple measurement intervals. Those filters would again require an expert to tune the thresholds for the filter output and the number of samples within the moving window.

5.2 *Time Series Profile Based Anomaly Detection*

This is the normal Machine Learning based Anomaly Detection. The predictor learns the normal time series profile. An anomaly is detected from a significant difference between the predicted value and the actual (measured) value. In figure 2 the DL (Down Link) is same as FL (Forward Link) in U.S. Both of them mean the direction from the Network to the User.

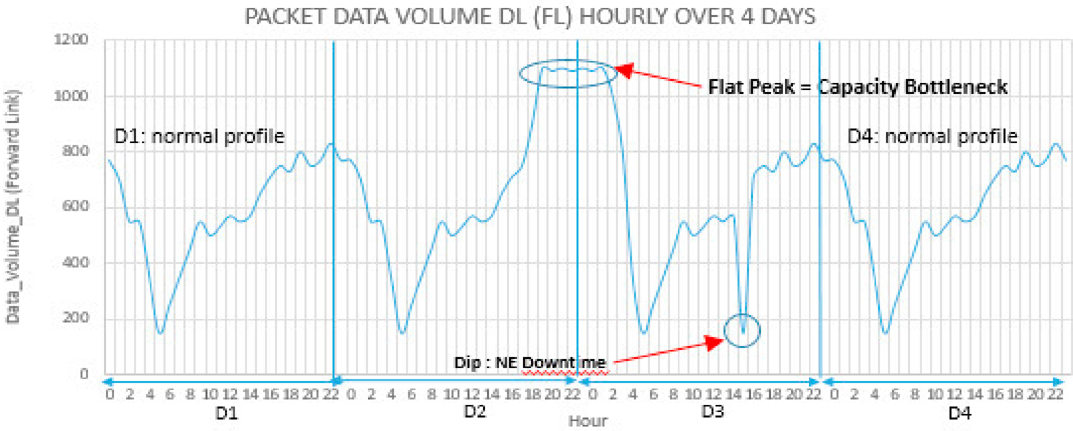


Figure 2: Example of daily profile based anomaly detection showing the hourly profile over 4 days

Pros: Detects volume (amount of something) based changes in time series profile, like:

- Total Packet Data Volume in Downlink (DL) / U.S. Forward Link (FL) Direction
- Total Packet Data Volume in Uplink (UL) / U.S. Reverse Link (RL) Direction

Cons: Does not understand (well) time series values that have e.g. a rate (ratio) based range like:

- Packet Data Session Setup Success Rate [%]
- Voice / Video Call Setup Success Rate (VOLTE, VOIP, Circuit Switched) [%]
- Incoming (or Outgoing) Handover / Handoff Failure Rate [%]
- Dropped (=Abnormally Released) Packet Data Session / Voice Call / Video Call Rate [%]
- Dropped Data Packet Rate [%]

This is because the rates tend to vary within a narrow range. In a typical system, the success rates tend to be close to 100% and the failure rates tend to be close to 0%. An exception can be processor load [%] that might vary e.g. 20% -70% in a server, with lots of users, creating a daily profile. However, processor load is not any success rate or failure rate type of KPI like the other rate examples above.

5.3 *Anomaly Filtering (AF)*

To reduce false detection we propose a combined approach requiring both of the above-mentioned conditions to be fulfilled. This means using both KPI Range Based AD and KPI Time Series Profile based AD. The required time series streams are available e.g. in advanced Cellular / Wireless / Telecommunications and any type of [Packet (e.g. IP)] Data / Computer Networks as well as in related network side / user side equipment.

Table 2 below contains an hourly time series profile over 24 days. An abnormal form of the daily profile produces an “Anomaly Candidate” for the (NE, Date, Hour) triplet in question. A subset of Anomaly Candidates is shown in **yellow color** in Table 2, but they are actually detected later from the **(Z>0)** values shown in Table 3.

Hour	**Day_1**	**…**	***Day_14***	**Day_15**	**…**	**Day_23**	**Day_24**	*Mean(Hour)*	*StdDev(Hour)*
0	801	…	945	876	…	1634	1494	*987*	*226*
…	…	…	…	…	…	…	…	…	…
4	220	…	***1127***	326	…	652	492	*554*	*229*
5	163	…	***1296***	276	…	893	319	*516*	*272*
…	…	…	…	…	…	…	…	…	…
23	865	…	1013	1154	…	1888	1741	*1102*	*282*

Table 2: The number of unwanted Events per day and hour based on real cellular /wireless network data

Following formula was adapted from [16]. It compares the Unwanted_Events (Day,Hour) to the Mean(Hour):

Z(Day, Hour) = Max(0 , ((Unwanted_Events(Day, Hour) – Mean(Hour)) / StdDev(Hour)) – StdDevFactor)

If **(Z(Day, Hour) > 0)** then **(Unwanted_Events(Day, Hour) – Mean(Hour)) > StdDevFactor * StdDev(Hour)**

Where ***StdDevFactor*** **(*>0*)** is a system parameter that can be set/modified by the user. In this study, we set the initial factor intuitively and we never saw a reason to change it.

Note: If (Unwanted_Events (Day, Hour) < Mean(Hour)) then 'it is good news', and Z=0

Hour	**Z(Day,Hour)**	**Day_1**	**…**	***Day_14***	**Day_15**	**…**	**Day_23**	**Day_24**
0	**Z(Day,0)**	0	…	0	0	…	0.9	0.2
…	**…**	…	…	…	…	…	…	…
4	**Z(Day,4)**	0	…	***0.5***	0	…	0	0
5	**Z(Day,5)**	0	…	***0.9***	0	…	0	0
…	**…**	…	…	…	…	…	…	…
23	**Z(Day,23)**	0	…	0	0	…	0.8	0.3

Table 3: Daily profile based anomaly candidates have (Z > 0), totally 35 cases, a subset is shown in yellow color.

An acceptable KPI range has been used for filtering in table 4. Normally the Carrier (U.S.) / The Network Operator defines this range. The range should be based on the current Quality of the Network and the number of fault cases the staff can daily handle with their existing tools and processes. If the KPI value is within the acceptable limits then any profile based Anomaly Candidate is ignored. The NE performance is still on an acceptable level relative to the rest of the network. No Field Engineers will be sent to the site. No brainpower will be wasted on Root Cause Analysis (RCA)**.**

Acceptable range in the Table 4: ***(0 <= KPI <= 2)*** The KPI value is the left-most decimal value in each slot

Hour	**Day_1**	**…**	***Day_14***	**…**	**Day_23**	**Day_24**
…	…	…	…	…	…	…
4	(1.6 <= 2) and *(Z=0)*	…	***(3.2 > 2) and (Z>0)***	…	**(2.1 > 2**) and *(Z=0)*	**(2.8 > 2**) and *(Z=0)*
5	(1.5 <= 2) and *(Z=0)*	…	***(4.2 > 2) and (Z>0)***	…	**(2.1 > 2**) and *(Z=0)*	**(2.3 > 2)** and *(Z=0)*
…	…	…	…	…	…	…

Table 4: KPI and Z(Day, Hour) : Only 2 Final Anomalies fulfil condition ((KPI out of [0,2] range) and (Z>0))

6. Machine Learning Module Implementation for the Proposed Method

We searched and studied anomaly detection papers for Cellular/ Wireless networks. We did not find any better efficient way to apply Supervised Machine Learning (SML) in AD using network generated big data [8,10,12].
As manual labelling of the training data is a major time-consuming step in the SML process, we looked for ways to automate it. Our proposed AF based automated labelling over-performed both the accuracy and speed of manual labelling in training set generation for SML.

Our final module, ML is fed with new labels produced by AF. We have measured 98.82% accuracy for test data sets. In this part we used KNN (K-nearest neighbors) algorithm with 3 nearest neighbors considered.

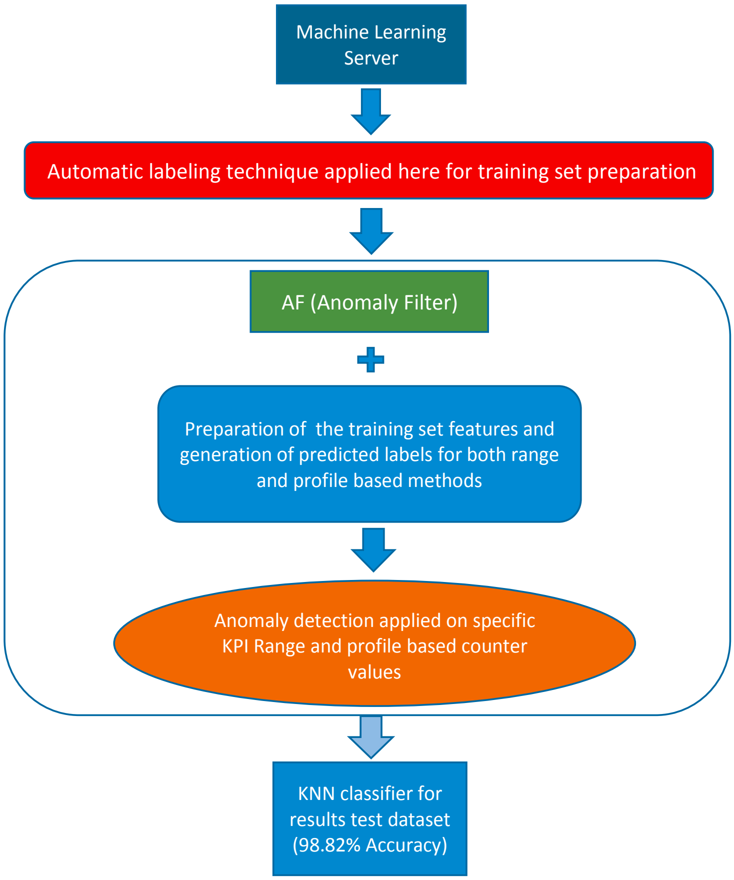
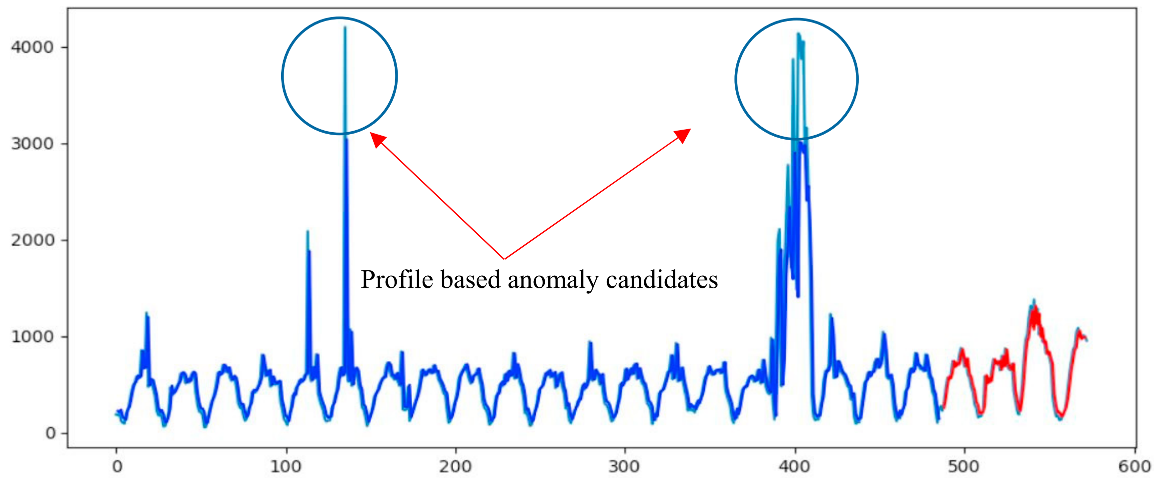


Figure 3: Machine Learning module implementation for automatic labelling of the training dataset

In this study, we have observed that it is hard to train a ML system with real network data, because anomalies are so rare events in the total amount of data. Therefore, our mathematical formulation based anomaly criteria, AF, proved very useful component for SML. At the same the automatic labelling saved us a lot of time and energy.

Figure 4 below graphs are from both ML module (a) and from an unsupervised ML algorithm generated results (b)

(a)



(b)

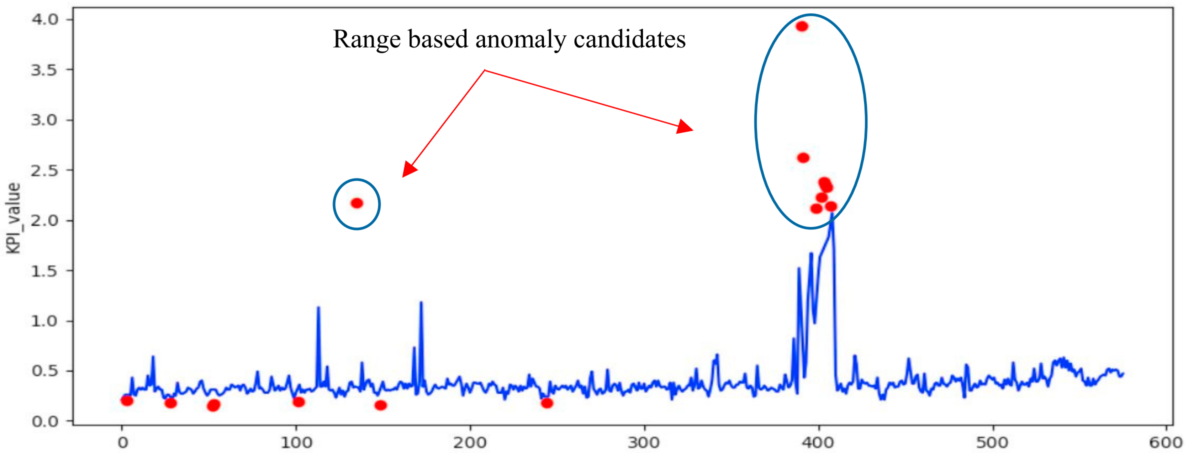


Figure 4: Machine learning module generated anomalies by both methods (a) LSTM-RNN based KPI profile prediction (b) SVM based KPI range prediction (in both graphs x axis shows system generated time tags)

In the figure 4, we are showing our results for unsupervised machine learning algorithms. We have applied one class support vector machine for range based KPI values. In our test, it found out the outliers of any used data set. We optimized the best-fit gamma parameter. The weak point of using unsupervised learning, it labels more FP or FN anomalies. This can be seen in the figure 4 (b) and marked with red dots. We decided to apply another LSTM (Long Short Term Memory) based RNN (Recurrent Neural Network) to predict the profile based KPI values in 4 (a). The real data is shown in light blue color and our model based prediction is shown in dark blue color.

Finally, we have proposed our combined model for SML. The combined model uses both anomaly predictions from range based and from profile based KPI values.

We assume AF to decrease the total time from the point of start of a fault to the point of corrective action completed, by a human or the by the system itself. This can be e.g. replacement of a faulty HW unit by a new / spare unit by a field engineer (no redundant unit installed or no remote control possibility). It might complement the rules to trigger an automatic switchover between redundant hot stand by units (e.g. N+N redundancy) and / or to trigger the warm-up followed by a switchover in case of a redundant cold standby unit (e.g. N+1 redundancy). This could improve the user experienced availability indirectly using KPIs to complete a lack in the self-diagnostics (Hiding Fault without Alarm), assuming a redundant unit is available.

We think our approach could be applied to Automation in general including Medical / any critical system or equipment, Telecommunications / Data Communication Networks extending to related network side / user side equipment. Both automated labelling of the training data set and the decrease of false positive anomalies are expected to speed up finding the true positive anomalies.

AD Method	False Positive (FP)	True Positive (TP)	True Positive Rate [%]
Acceptable Range	36	2	6%
Profile Based	33	2	5%
Anomaly Filtering	**0**	**2**	**100%**

Table 5: Performance of Anomaly Detection methods in our study with real Cellular / Wireless Network KPI data

7. Conclusions

Based on this study our enhanced proposal for AD, AF ,decreased the number of FP detections from 33 to 0. The total number of anomalies decreased from 35 uncertain cases down to 2 TP. There were no FN cases before / after AF. Our conclusion is that using AF, firstly time series profile based KPI AD is followed by acceptable KPI Range based AD (or vice versa), can provide a major improvement to the accuracy. The required input data time series streams are available in advanced Cellular / Wireless and other Telecommunications Networks as well as in Computer / [Packet (IP)] Data based Networks and within related Network Elements, Handsets and other Equipment. See [15] and [17] for examples of Performance Management (PM) raw counters and KPI formula. We assume our approach can also save time in System Analysis and Diagnostics of e.g. Automation, Telecommunications and Data Communications Networks, extending to the related network side / user side equipment. Our SML proposal is able to reduce the FP and FN anomalies. Test results support our idea and we are hopeful to apply this novel approach to new application domains as an AD algorithm.

8. Acknowledgements

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